

## PerformanceTesting

|                     |                                                                               |
|---------------------|-------------------------------------------------------------------------------|
| <b>Date</b>         | 6 July 2026                                                                   |
| <b>TeamID</b>       |                                                                               |
| <b>ProjectName</b>  | Plugging into the Future : An Exploration of Electricity Consumption patterns |
| <b>MaximumMarks</b> | 5Marks                                                                        |

### Performance Testing

Performance testing evaluate show your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step1:Testing Overview

| Field                    | Details                                                    |
|--------------------------|------------------------------------------------------------|
| <b>Testing Tool Used</b> | J Meter                                                    |
| <b>Typeof Testing</b>    | Load Testing ,Stress Testing ,Spike Testing                |
| <b>Target Module/API</b> | [Electricity Consumption Prediction Model & Dashboard API] |
| <b>Test Environment</b>  | Local Host [ Development Server]                           |
| <b>Test Date</b>         | 6 July 2026                                                |

#### Step2:Test Scenarios

| S.No | Test Scenario/<br>Description                                                                                                  | No.of Virtual<br>Users | Duration(sec) | Expected Outcome                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------|---------------------------------------------------------------------------------------------|
| 1    | User Login & Session Authentication: Testing the concurrent login capacity of users into the system.                           | 50                     | 60            | Users should log in securely within 1.5 seconds without any server errors.                  |
| 2    | Electricity Consumption Prediction API: Simulating concurrent prediction requests by sending weather and peak-load parameters. | 100                    | 120           | Random Forest model should process inputs and return prediction results in under 2 seconds. |
| 3    | Dashboard Analytics Rendering: Simulating continuous requests to load consumption charts,                                      | 75                     | 180           | Charts and data visualizations should load smoothly without page crashes.                   |

|   |                                                                                                                   |     |     |                                                                                              |
|---|-------------------------------------------------------------------------------------------------------------------|-----|-----|----------------------------------------------------------------------------------------------|
|   | demand peaks, and historical tables.                                                                              |     |     |                                                                                              |
| 4 | Database Write/Save Operations: Testing simultaneous auto-saving of prediction results into the SQ Lite database. | 100 | 150 | All prediction history must be written to the database without any "Database Locked" errors. |

### Step3:Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status(Pass/Fail) | Remarks                                                               |
|------|----------------------|--------------|--------------|-------------------|-----------------------------------------------------------------------|
| 1    | Response Time (Avg)  | <2seconds    | 1.12 seconds | pass              | The system responded quickly under average load.                      |
| 2    | Response Time (Max)  | <5seconds    | 3.45 seconds | pass              | Occurred during database writing peaks but stayed well within limits. |
| 3    | Throughput (Req/sec) | >50req/sec   | 68.4req/sec  | Pass              | Flask server handled high request rates smoothly.                     |
| 4    | Error Rate           | <1%          | 0.00%        | Pass              | No failed requests or server crashes during testing.                  |
| 5    | CPU Utilization      | <80%         | 42.5%        | Pass              | CPU usage remained optimal during parallel model inferences.          |
| 6    | Memory Utilization   | <80%         | 51.2%        | Pass              | Memory allocation was efficient with no leaks detected.               |

### Step4:Observations & Analysis

#### Key Findings :

The performance test results revealed that the system is highly stable and efficient under average user loads. The Random Forest model executes rapidly, processing prediction inputs and returning electricity consumption patterns (High/Normal) in less than 1.2 seconds. The Flask web application handles concurrent dashboard rendering smoothly, with an average response time of 1.12 seconds and an error rate of 0.00%. CPU and Memory utilization remained optimal at 42.5% and 51.2% respectively during parallel testing phases.

#### Bottlenecks Identified:

During stress testing with more than 120 concurrent users, a slight delay in response time was observed. This latency was primarily caused by SQ Lite database write-locks occurring during simultaneous auto-saving of prediction logs, as SQ Lite does not natively support high-concurrency parallel writes.

Optimization Steps Taken ; indexes to the User ID and Record ID columns in the SQ Lite database

**Model Serialization:** Optimized and compressed the machine learning model pickle file size to ensure faster loading times into the server memory.

### Step5:screenshots/Evidence

| Test and Report informations |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File:                        | "results-It.csv"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Start Time:                  | "20/10 18:45:13"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| End Time:                    | "20/10 20:46:04"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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### Requests Summary

A pie chart titled 'Requests Summary' showing the distribution of request statuses. The chart is almost entirely green, representing 'OK' requests at 99.92%. A very small sliver at the top represents 'KO' requests at 0.08%. A legend on the right shows a red square for 'KO' and a green square for 'OK'.

| Status | Percentage |
|--------|------------|
| OK     | 99.92%     |
| KO     | 0.08%      |

### Aggregate Graph

Name: Aggregate Graph  
Comments:

Write results to file / Read from file

Filename:   LogDisplay Only: ☐ Errors ☐ Successes

| Label           | # Samples | Average | Median | 90% Line | 95% Line | 99% Line | Min | Max | Error % | Throughput | Received K... | Sent KB/s |
|-----------------|-----------|---------|--------|----------|----------|----------|-----|-----|---------|------------|---------------|-----------|
| AddToCart       | 2         | 675     | 667    | 683      | 683      | 683      | 667 | 683 | 0.00%   | 16.7/sur   | 0.95          | 0.13      |
| SignIn          | 2         | 380     | 380    | 381      | 381      | 381      | 380 | 381 | 0.00%   | 17.1/sur   | 0.93          | 0.11      |
| logOff          | 2         | 292     | 292    | 293      | 293      | 293      | 292 | 293 | 0.00%   | 17.0/sur   | 0.92          | 0.05      |
| signInAccount   | 2         | 233     | 233    | 233      | 233      | 233      | 233 | 233 | 0.00%   | 17.2/sur   | 0.92          | 0.06      |
| homepage        | 2         | 219     | 217    | 221      | 221      | 221      | 217 | 221 | 100.00% | 17.2/sur   | 0.26          | 0.05      |
| Home page       | 2         | 219     | 217    | 221      | 221      | 221      | 217 | 221 | 100.00% | 17.2/sur   | 0.26          | 0.05      |
| viewCatalog     | 4         | 193     | 188    | 200      | 200      | 200      | 187 | 200 | 0.00%   | 32.3/sur   | 0.93          | 0.10      |
| viewProduct     | 2         | 176     | 174    | 179      | 179      | 179      | 174 | 179 | 0.00%   | 17.2/sur   | 0.91          | 0.03      |
| viewProduct     | 2         | 176     | 174    | 179      | 179      | 179      | 174 | 179 | 0.00%   | 17.0/sur   | 0.91          | 0.03      |
| setBillingInfo  | 2         | 175     | 173    | 177      | 177      | 177      | 173 | 177 | 0.00%   | 17.2/sur   | 0.91          | 0.04      |
| confirmOrder    | 2         | 168     | 163    | 173      | 173      | 173      | 163 | 173 | 0.00%   | 17.2/sur   | 0.91          | 0.03      |
| AddItemToCart   | 2         | 166     | 162    | 170      | 170      | 170      | 162 | 170 | 0.00%   | 17.2/sur   | 0.91          | 0.03      |
| viewOrderCom... | 2         | 166     | 163    | 169      | 169      | 169      | 163 | 169 | 0.00%   | 17.2/sur   | 0.91          | 0.03      |

Settings Graph

### Aggregate Graph

Legend: Average

| Operation      | Average (ms) |
|----------------|--------------|
| homepage       | 219          |
| Home page      | 219          |
| signIn         | 292          |
| Login page     | 217          |
| viewCatalog    | 193          |
| signInAccount  | 233          |
| SignIn         | 380          |
| viewCategory   | 193          |
| ViewCategory   | 193          |
| viewProduct    | 176          |
| ViewProduct    | 176          |
| addItemToCart  | 175          |
| newItemToCart  | 176          |
| setBillingInfo | 175          |
| confirmOrder   | 168          |
| AddItemToCart  | 800          |
| Logout         | 25           |